NUTAN MAHARASHTRA INSTITUTE OF ENGINEERING & TECHNOLOGY

Department of Computer Engineering

LABORATORY MANUAL & OUTPUT RECORD

Object Oriented Programming Laboratory (CE25PCC-205)

Student Name:	Rohit Sandeep Todkar
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EXP 3	**Publication Management & Exception Handling** `EXP3.cpp`	Single Inheritance, Exception Handling (try-throw-catch), Robust Input Validation	Completed
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EXP 1

EXP1.cpp

Complex Number Operations using Operator Overloading

> **Problem Statement:** Implement a class Complex which represents the Complex Number data type. Implement constructors (including default constructor for 0+0i). Overload operator+ to add two complex numbers, operator* to multiply two complex numbers, and operators << and >> to print and read Complex numbers.
> **Core Concepts:** Classes, Constructors, Operator Overloading (+, *, >>, <<), Friend Functions

SOURCE CODE:

```
// Experiment 1: Complex Number Operations using Operator Overloading
#include <iostream>
using namespace std;

class Complex
{
private:
    int real;
    int imag;

public:
    // Default Constructor
    Complex()
    {
        real = 0;
        imag = 0;
    }

    // Parameterized Constructor
    Complex(int r, int i)
    {
        real = r;
        imag = i;
    }

    // Overload + operator
    Complex operator +(Complex c)
    {
        Complex temp;
        temp.real = real + c.real;
        temp.imag = imag + c.imag;
        return temp;
    }

    // Overload * operator
    Complex operator *(Complex c)
    {
        Complex temp;
        temp.real = (real * c.real) - (imag * c.imag);
        temp.imag = (real * c.imag) + (imag * c.real);
        return temp;
    }

    // Overload >> operator
    friend istream& operator >>(istream& in, Complex& c);

    // Overload << operator
    friend ostream& operator <<(ostream& out, Complex& c);
};
```

```
// Definition of >> operator
istream& operator >>(istream& in, Complex& c)
{
    cout << "Enter Real Part : ";
    in >> c.real;

    cout << "Enter Imaginary Part : ";
    in >> c.imag;

    return in;
}

// Definition of << operator
ostream& operator <<(ostream& out, Complex& c)
{
    out << c.real;

    if (c.imag ≥ 0)
        out << "+" << c.imag << "i";
    else
        out << c.imag << "i";

    return out;
}

// Main Function
int main()
{
    Complex c1, c2, sum, product;

    cout << "Enter First Complex Number" << endl;
    cin >> c1;

    cout << "\nEnter Second Complex Number" << endl;
    cin >> c2;

    sum = c1 + c2;
    product = c1 * c2;

    cout << "\nFirst Complex Number = " << c1 << endl;
    cout << "Second Complex Number = " << c2 << endl;

    cout << "\nAddition = " << sum << endl;
    cout << "Multiplication = " << product << endl;

    return 0;
}
```

TEST INPUT:

```
Enter First Complex Number: 3 4
Enter Second Complex Number: 1 2
```

EXECUTION OUTPUT:

```
Terminal — ./a.out

Enter First Complex Number
Enter Real Part : Enter Imaginary Part :
Enter Second Complex Number
Enter Real Part : Enter Imaginary Part :
```

```
First Complex Number = 3+4i
Second Complex Number = 1+2i

Addition = 4+6i
Multiplication = -5+10i
```

EXP 2

EXP2.cpp

Staff Payroll Management & Sorting

Problem Statement: Create a class staff having fields: Staff-id, name, salary and functions accept() and display(). Calculate DA=25% of basic salary, HRA=800, I-tax=15% of basic salary. Display the pay slip using appropriate output format. Sort the data by name.
Core Concepts: Classes and Objects, Array of Objects, Encapsulation, Sorting Algorithm

SOURCE CODE:

```
// Experiment 2: Staff Payroll Management & Sorting
#include<iostream>
#include<string>

using namespace std;


class staff
{
private:
  int Staff_ID;
  string name;
  float Basic_Salary;
  float DA, HRA, Itax, NetSalary;

public:

  void accept()
  {
    cout<<"\nEnter staff ID : \t";
    cin>> Staff_ID;
    cout<<"\nEnter staff Name : \t";
    cin>> name;
    cout<<"\nEnter staff Salary : \t";
    cin>> Basic_Salary;
  }//accept

  void calculate()
  {
    DA = Basic_Salary * 0.25 ;
    HRA = 800;
    Itax = Basic_Salary *0.15;
    NetSalary= Basic_Salary + DA + HRA -Itax;
  }//calc

  void display()
  {
    cout<<"\n\n\nStaff ID \t"<< Staff_ID;
    cout<<"\nStaff Name \t"<< name;
    cout<<"\nBasic Salary \t"<< Basic_Salary;
    cout<<"\nDA(25%) \t"<< DA;
    cout<<"\nHRA \t"<< HRA;
    cout<<"\nIncome Tax\t"<< Itax;
    cout<<"\nNet Salary \t"<< NetSalary << endl;

  }//display

  string getName()
  {
```

```
    return name;
  } //name

}; // class staff

int main()
{
  int i,j,n;

  cout<<"Enter Number of staff you want to fill\t";
  cin >> n;
  staff s[n];

  for(i=0;i<n;i++)
  {
    cout<<"\nEnter staff Details of staff "<< i +1 << endl;
    s[i].accept();
    s[i].calculate();
  }//for loop Accept data

  for(i=0;i<n-1;i++)
  {
    for(j=i+1;j<n;j++)
    {
      if(s[i].getName()>s[j].getName())
      {
        swap(s[i],s[j]);
      }
    }
  }//for loop sorting

  cout<<"\n staff pay slip";

  for(i=0;i<n;i++)
  {
    s[i].display();
  }//for loop for display

  return 0;


}//main
```

TEST INPUT:

```
Enter Number of staff: 2
Staff 1: ID = 101, Name = Alice, Basic = 50000
Staff 2: ID = 102, Name = Bob, Basic = 60000
```

EXECUTION OUTPUT:

```
Terminal — ./a.out

Enter Number of staff you want to fill
Enter staff Details of staff 1
Enter staff ID :
Enter staff Name :
Enter staff Salary :
Enter staff Details of staff 2
Enter staff ID :
Enter staff Name :
Enter staff Salary :
```

```
 staff pay slip

Staff ID        101
Staff Name      Alice
Basic Salary    50000
DA(25%)         12500
HRA     800
Income Tax      7500
Net Salary      55800

Staff ID        102
Staff Name      Bob
Basic Salary    60000
DA(25%)         15000
HRA     800
Income Tax      9000
Net Salary      66800
```

EXP 3

EXP3.cpp

Publication Management & Exception Handling

Problem Statement: Imagine a publishing company which does marketing for book and audio cassette versions. Create a class publication that stores title and price. Derive two classes: book (page count) and disc/tape (playing time). Catch exceptions for invalid entries and reset to zero.
Core Concepts: Single Inheritance, Exception Handling (try-throw-catch), Robust Input Validation

SOURCE CODE:

```
// Experiment 3: Publication Management & Exception Handling
#include<iostream>
#include<string>
using namespace std;

class publications
{
protected:
    float price;
    string title;

public:
    void details()
    {
        try
        {
            cout << "Enter the title: ";
            cin >> ws;
            getline(cin, title);

            cout << "Enter price: ";
            cin >> price;

            if(price < 0)
                throw price;
        }
        catch( ... )
        {
            cout << "Invalid price entered!" << endl;
            setting_zero();
        }
    }

    void display()
    {
        cout << "Title : " << title << endl;
        cout << "Price : " << price << endl;
    }

    void setting_zero()
    {
        title = "Invalid";
        price = 0;
    }
};

class book : public publications
{
private:
```

```
    int pagecount;

public:
    void details_book()
    {
        details();

        try
        {
            cout << "Enter page count: ";
            cin >> pagecount;

            if(pagecount < 0)
                throw pagecount;
        }
        catch( ... )
        {
            cout << "Invalid page count entered!" << endl;
            pagecount = 0;
        }
    }

    void display_book()
    {
        cout << "\n********** BOOK DETAILS **********" << endl;
        display();
        cout << "Page Count : " << pagecount << endl;
    }
};

class disc : public publications
{
private:
    int time;

public:
    void details_disc()
    {
        details();

        try
        {
            cout << "Enter playing time (minutes): ";
            cin >> time;

            if(time < 0)
                throw time;
        }
        catch( ... )
        {
            cout << "Invalid playing time entered!" << endl;
            time = 0;
        }
    }

    void display_disc()
    {
        cout << "\n********** DISC DETAILS **********" << endl;
        display();
        cout << "Playing Time : " << time << " minutes" << endl;
    }
};

int main()
{
    book b;
    disc d;
```

```
    cout << "Enter Book Details" << endl;
    b.details_book();

    cout << "\nEnter Disc Details" << endl;
    d.details_disc();

    cout << "\nDisplaying Details\n";

    b.display_book();
    d.display_disc();

    return 0;
}
```

TEST INPUT:

```
Book: Title = C++ Primer, Price = 450.50, Pages = 800
Disc: Title = Mastering Audio, Price = 299.99, Time = 60 min
```

EXECUTION OUTPUT:

```
Terminal — ./a.out

Enter Book Details
Enter the title: Enter price: Enter page count:

Enter Disc Details
Enter the title: Enter price: Enter playing time (minutes):

Displaying Details

********** BOOK DETAILS **********
Title : C++ Primer
Price : 450.5
Page Count : 800

********** DISC DETAILS **********
Title : Mastering Audio
Price : 299.99
Playing Time : 60 minutes
```

EXP 4

EXP4.cpp

Matrix ADT Operations (Addition, Subtraction, Multiplication)

Problem Statement: Implement the Matrix Abstract Data Type (ADT) using a class. Supported operations: Reading a matrix, printing a matrix, addition of matrices, subtraction of matrices, and multiplication of matrices with dimension validation.
Core Concepts: Matrix ADT, 2D Arrays in Classes, Object Parameter Passing & Return, Algebra Checks

SOURCE CODE:

```
// Experiment 4: Matrix ADT Operations (Addition, Subtraction, Multiplication)
#include <iostream>
using namespace std;

class Matrix
{
private:
    int row, col;
    int a[10][10];

public:
    void readMatrix()
    {
        cout << "Enter number of rows and columns: ";
        cin >> row >> col;

        cout << "Enter matrix elements:\n";
        for (int i = 0; i < row; i++)
        {
            for (int j = 0; j < col; j++)
            {
                cin >> a[i][j];
            }
        }
    }

    void displayMatrix()
    {
        for (int i = 0; i < row; i++)
        {
            for (int j = 0; j < col; j++)
            {
                cout << a[i][j] << "\t";
            }
            cout << endl;
        }
    }

    Matrix addMatrix(Matrix m)
    {
        Matrix temp;

        if (row != m.row || col != m.col)
        {
            cout << "Addition not possible.\n";
            temp.row = temp.col = 0;
            return temp;
        }
```

```
        temp.row = row;
        temp.col = col;

        for (int i = 0; i < row; i++)
        {
            for (int j = 0; j < col; j++)
            {
                temp.a[i][j] = a[i][j] + m.a[i][j];
            }
        }

        return temp;
    }

    Matrix subtractMatrix(Matrix m)
    {
        Matrix temp;

        if (row ≠ m.row || col ≠ m.col)
        {
            cout << "Subtraction not possible.\n";
            temp.row = temp.col = 0;
            return temp;
        }

        temp.row = row;
        temp.col = col;

        for (int i = 0; i < row; i++)
        {
            for (int j = 0; j < col; j++)
            {
                temp.a[i][j] = a[i][j] - m.a[i][j];
            }
        }

        return temp;
    }

    Matrix multiplyMatrix(Matrix m)
    {
        Matrix temp;

        if (col ≠ m.row)
        {
            cout << "Multiplication not possible.\n";
            temp.row = temp.col = 0;
            return temp;
        }

        temp.row = row;
        temp.col = m.col;

        for (int i = 0; i < temp.row; i++)
        {
            for (int j = 0; j < temp.col; j++)
            {
                temp.a[i][j] = 0;

                for (int k = 0; k < col; k++)
                {
                    temp.a[i][j] += a[i][k] * m.a[k][j];
                }
            }
        }

        return temp;
    }
```

```
    bool isValid()
    {
        return (row > 0 && col > 0);
    }
};

int main()
{
    Matrix A, B, C;

    cout << "Enter First Matrix\n";
    A.readMatrix();

    cout << "\nEnter Second Matrix\n";
    B.readMatrix();

    cout << "\nFirst Matrix:\n";
    A.displayMatrix();

    cout << "\nSecond Matrix:\n";
    B.displayMatrix();

    cout << "\nAddition of Matrices:\n";
    C = A.addMatrix(B);
    if (C.isValid())
        C.displayMatrix();

    cout << "\nSubtraction of Matrices:\n";
    C = A.subtractMatrix(B);
    if (C.isValid())
        C.displayMatrix();

    cout << "\nMultiplication of Matrices:\n";
    C = A.multiplyMatrix(B);
    if (C.isValid())
        C.displayMatrix();

    return 0;
}
```

TEST INPUT:

```
Matrix A (2x2): [[1, 2], [3, 4]]
Matrix B (2x2): [[5, 6], [7, 8]]
```

EXECUTION OUTPUT:

```
Terminal — ./a.out

Enter First Matrix
Enter number of rows and columns: Enter matrix elements:

Enter Second Matrix
Enter number of rows and columns: Enter matrix elements:

First Matrix:
1       2
3       4

Second Matrix:
5       6
7       8
```

```
Addition of Matrices:
6       8
10      12

Subtraction of Matrices:
-4      -4
-4      -4

Multiplication of Matrices:
19      22
43      50
```

EXP 5

EXP5.cpp

Constructor & Destructor Order in Multiple Inheritance

> **Problem Statement:** Illustrate default constructor, parameterized constructor, and copy constructor. Demonstrate the order of execution of constructors and destructors when a new class (Result) is derived from more than one base class (Student and Sports).
> **Core Concepts:** Multiple Inheritance, Default/Parameterized/Copy Constructors, Destructor Invocation Order

SOURCE CODE:

```
// Experiment 5: Constructor & Destructor Order in Multiple Inheritance
#include<iostream>
using namespace std;
class Student {
protected:
    int rollNo;
public:
    Student() {
        rollNo = 0;
        cout << "Student Default Constructor Called" << endl;
    }
    Student(int r) {
        rollNo = r;
        cout << "Student Parameterized Constructor Called" << endl;
    }
    Student(const Student &s) {
        rollNo = s.rollNo;
        cout << "Student Copy Constructor Called" << endl;
    }
    ~Student() {
        cout << "Student Destructor Called" << endl;
    }
};

class Sports {
protected:
    int sportsMarks;
public:
    Sports() {
        sportsMarks = 0;
        cout << "Sports Default Constructor Called" << endl;
    }
    Sports(int m) {
        sportsMarks = m;
        cout << "Sports Parameterized Constructor Called" << endl;
    }
    Sports(const Sports &s) {
        sportsMarks = s.sportsMarks;
        cout << "Sports Copy Constructor Called" << endl;
    }
    ~Sports() {
        cout << "Sports Destructor Called" << endl;
    }
};


class Result : public Student, public Sports {
private:
    int totalMarks;
```

```
public:
    Result() : Student(), Sports() {
        totalMarks = 0;
        cout << "Result Default Constructor Called" << endl;
    }
    Result(int r, int s, int t) : Student(r), Sports(s) {
        totalMarks = t;
        cout << "Result Parameterized Constructor Called" << endl;
    }
    Result(const Result &obj) : Student(obj), Sports(obj) {
        totalMarks = obj.totalMarks;
        cout << "Result Copy Constructor Called" << endl;
    }
    void display() {
        cout << "\nRoll No : " << rollNo;
        cout << "\nSports Marks : " << sportsMarks;
        cout << "\nTotal Marks : " << totalMarks << endl;
    }
    ~Result() {
        cout << "Result Destructor Called" << endl;
    }
};


int main() {
    cout << "\n===== Default Constructor =====\n";
    Result r1;
    r1.display();

    cout << "\n===== Parameterized Constructor =====\n";
    Result r2(101, 25, 450);
    r2.display();

    cout << "\n===== Copy Constructor =====\n";
    Result r3(r2);
    r3.display();

    cout << "\nProgram Ends..." << endl;
    return 0;
}
```

TEST INPUT:

> None (Demonstrated programmatically in main)

EXECUTION OUTPUT:

```
Terminal — ./a.out

===== Default Constructor =====
Student Default Constructor Called
Sports Default Constructor Called
Result Default Constructor Called

Roll No : 0
Sports Marks : 0
Total Marks : 0

===== Parameterized Constructor =====
Student Parameterized Constructor Called
Sports Parameterized Constructor Called
Result Parameterized Constructor Called
```

```
Roll No : 101
Sports Marks : 25
Total Marks : 450

===== Copy Constructor =====
Student Copy Constructor Called
Sports Copy Constructor Called
Result Copy Constructor Called

Roll No : 101
Sports Marks : 25
Total Marks : 450

Program Ends ...
Result Destructor Called
Sports Destructor Called
Student Destructor Called
Result Destructor Called
Sports Destructor Called
Student Destructor Called
Result Destructor Called
Sports Destructor Called
Student Destructor Called
```

EXP 6

EXP6.cpp

File Input and Output Operations

> **Problem Statement:** Create an output file, write information into the file, close it, then reopen the same file as an input file and read the stored information for display.
> **Core Concepts:** File Streams (ofstream, ifstream), File Creation, Writing, Closing, Reopening & Reading

SOURCE CODE:

```
// Experiment 6: File Input and Output Operations
#include <iostream>
#include <fstream>
#include <string>

int main() {
    const std::string filename = "sample_output.txt";

    std::ofstream outFile(filename);

    if (!outFile.is_open()) {
        std::cerr << "Error: Could not create or open the output file." << std::endl;
        return 1;
    }

    std::cout << "Writing data to '" << filename << "'...\n";
    outFile << "Line 1: Hello, C++ File I/O!\n";
    outFile << "Line 2: Demonstration of writing to a file.\n";
    outFile << "Line 3: Storing integer value = " << 100 << "\n";
    outFile << "Line 4: NMIET " << "\n";


    outFile.close();
    std::cout << "File closed successfully after writing.\n\n";

    std::ifstream inFile(filename);

    if (!inFile.is_open()) {
        std::cerr << "Error: Could not open the file for reading." << std::endl;
        return 1;
    }

    std::cout << "--- Contents of '" << filename << "' ---" << std::endl;

    std::string line;
    while (std::getline(inFile, line)) {
        std::cout << line << std::endl;
    }

    inFile.close();
    std::cout << "-------------------------------------------------" << std::endl;

    return 0;
}
```

TEST INPUT:

```
None (Demonstrated using sample_output.txt)
```

EXECUTION OUTPUT:

Terminal — ./a.out

```
Writing data to 'sample_output.txt' ...
File closed successfully after writing.

--- Contents of 'sample_output.txt' ---
Line 1: Hello, C++ File I/O!
Line 2: Demonstration of writing to a file.
Line 3: Storing integer value = 100
Line 4: NMIET
------------------------------------------------
```

EXP 7

EXP7.cpp

Types of Inheritance (Single, Multiple, Multilevel, Hierarchical)

Problem Statement: Illustrate how the following forms of inheritance are supported in C++: a. Single inheritance, b. Multiple inheritance, c. Multilevel inheritance, d. Hierarchical inheritance.
Core Concepts: Inheritance Types, Base & Derived Classes, Function Overriding, Hierarchy Organization

SOURCE CODE:

```
// Experiment 7: Types of Inheritance (Single, Multiple, Multilevel, Hierarchical)
#include <iostream>
using namespace std;

// 1. Single Inheritance (7-1- single)
namespace SingleInheritance {
class Student {
public:
  void getStudent() { cout << "Student details" << endl; }
};

class Result : public Student {
public:
  void displayResult() { cout << "Result: Pass" << endl; }
};

void run() {
  Result r;
  r.getStudent();    // Base class function
  r.displayResult(); // Derived class function
}
} // namespace SingleInheritance

// 2. Multiple Inheritance (7-2- multiple)
namespace MultipleInheritance {
class Academic {
public:
  void academicMarks() { cout << "Academic Marks: 80" << endl; }
};

class Sports {
public:
  void sportsMarks() { cout << "Sports Marks: 15" << endl; }
};

class Student : public Academic, public Sports {
public:
  void display() { cout << "Total Performance: Good" << endl; }
};

void run() {
  Student s;
  s.academicMarks();
  s.sportsMarks();
  s.display();
}
} // namespace MultipleInheritance

// 3. Multilevel Inheritance (7-3- multilevel)
namespace MultilevelInheritance {
```

```
class Person {
public:
  void showPerson() { cout << "I am a Person" << endl; }
};

class Student : public Person {
public:
  void showStudent() { cout << "I am a Student" << endl; }
};

class Result : public Student {
public:
  void showResult() { cout << "Result: Pass" << endl; }
};

void run() {
  Result r;
  r.showPerson();  // Person class
  r.showStudent(); // Student class
  r.showResult();  // Result class
}
} // namespace MultilevelInheritance

// 4. Hierarchical Inheritance (7-4- hierarchical)
namespace HierarchicalInheritance {
class Employee {
public:
  void showEmployee() { cout << "Employee ID: 101" << endl; }
};

class Developer : public Employee {
public:
  void showDeveloper() { cout << "Role: Software Developer" << endl; }
};

class Manager : public Employee {
public:
  void showManager() { cout << "Role: Project Manager" << endl; }
};

void run() {
  Developer d;
  Manager m;

  cout << "Developer Details:" << endl;
  d.showEmployee();
  d.showDeveloper();

  cout << "\nManager Details:" << endl;
  m.showEmployee();
  m.showManager();
}
} // namespace HierarchicalInheritance

int main() {
  cout << "=== 1. Single Inheritance ===" << endl;
  SingleInheritance::run();

  cout << "\n=== 2. Multiple Inheritance ===" << endl;
  MultipleInheritance::run();

  cout << "\n=== 3. Multilevel Inheritance ===" << endl;
  MultilevelInheritance::run();

  cout << "\n=== 4. Hierarchical Inheritance ===" << endl;
  HierarchicalInheritance::run();
```

```
    return 0;
}
```

TEST INPUT:

> None (Demonstrated programmatically in main)

EXECUTION OUTPUT:

Terminal — ./a.out

```
=== 1. Single Inheritance ===
Student details
Result: Pass

=== 2. Multiple Inheritance ===
Academic Marks: 80
Sports Marks: 15
Total Performance: Good

=== 3. Multilevel Inheritance ===
I am a Person
I am a Student
Result: Pass

=== 4. Hierarchical Inheritance ===
Developer Details:
Employee ID: 101
Role: Software Developer

Manager Details:
Employee ID: 101
Role: Project Manager
```

EXP 8

EXP8.cpp

Generic Library Management System using Class Templates

> **Problem Statement:** Design and implement a generic Library Management System in C++ using Class Templates capable of handling integer as well as string-based book identification tags.
> **Core Concepts:** Class Templates, Generic Programming, Type-agnostic Member Functions

SOURCE CODE:

```
// Experiment 8: Generic Library Management System using Class Templates
#include <iostream>
#include <string>

using namespace std;

// Class Template
template <class T> class Library {
private:
  T bookID;
  string bookName;
  string author;
  bool issued;

public:
  // Constructor
  Library(T id, string name, string auth) {
    bookID = id;
    bookName = name;
    author = auth;
    issued = false;
  }

  // Display book details
  void display() {
    cout << "\nBook ID: " << bookID;
    cout << "\nBook Name: " << bookName;
    cout << "\nAuthor: " << author;
    cout << "\nStatus: ";
    if (issued)
      cout << "Issued";
    else
      cout << "Available";
    cout << "\n";
  }

  // Issue book
  void issueBook() {
    if (!issued) {
      issued = true;
      cout << "\nBook issued successfully.\n";
    } else {
      cout << "\nBook is already issued.\n";
    }
  }

  // Return book
  void returnBook() {
    if (issued) {
      issued = false;
```

```
      cout << "\nBook returned successfully.\n";
    } else {
      cout << "\nBook was not issued.\n";
    }
  }
};

int main() {
  // Library object with integer Book ID
  Library<int> book1(101, "Object Oriented Programming", "Robert Lafore");

  // Library object with string Book ID
  Library<string> book2("CS102", "Database Management System", "Korth");

  cout << "LIBRARY MANAGEMENT SYSTEM\n";

  cout << "\nBook 1 Details:";
  book1.display();

  cout << "\nIssuing Book 1:";
  book1.issueBook();

  cout << "\nBook 1 Details after issuing:";
  book1.display();

  cout << "\n--------------------------------\n";

  cout << "\nBook 2 Details:";
  book2.display();

  return 0;
}
```

TEST INPUT:

> None (Demonstrated programmatically in main)

EXECUTION OUTPUT:

Terminal — ./a.out

```
LIBRARY MANAGEMENT SYSTEM

Book 1 Details:
Book ID: 101
Book Name: Object Oriented Programming
Author: Robert Lafore
Status: Available

Issuing Book 1:
Book issued successfully.

Book 1 Details after issuing:
Book ID: 101
Book Name: Object Oriented Programming
Author: Robert Lafore
Status: Issued

--------------------------------

Book 2 Details:
Book ID: CS102
```

```
Book Name: Database Management System
Author: Korth
Status: Available
```

EXP 9

EXP9.cpp

State Population Lookup using STL Map

> **Problem Statement:** Use map associative container where keys are names of states and values are populations of states. Prompt user for state name, look up in map using state name as key, and return population.
> **Core Concepts:** Standard Template Library (STL), std::map Associative Container, find() Iterator Lookup

SOURCE CODE:

```
// Experiment 9: State Population Lookup using STL Map
#include <iostream>
#include <map>
#include <string>
using namespace std;

int main() {
  map<string, long long> statePopulation;

  statePopulation["Maharashtra"] = 126000000;
  statePopulation["Gujarat"] = 72000000;
  statePopulation["Karnataka"] = 68000000;
  statePopulation["Rajasthan"] = 82000000;
  statePopulation["Tamil Nadu"] = 77000000;
  statePopulation["Uttar Pradesh"] = 240000000;
  statePopulation["Madhya Pradesh"] = 88000000;

  string stateName;

  cout << "Enter the name of a state: ";
  getline(cin, stateName);

  map<string, long long>::iterator it = statePopulation.find(stateName);

  if (it != statePopulation.end()) {
    cout << "Population of " << stateName << " = " << it->second << endl;
  } else {
    cout << "State not found in the map." << endl;
  }

  return 0;
}
```

TEST INPUT:

```
Enter the name of a state: Maharashtra
```

EXECUTION OUTPUT:

```
Terminal — ./a.out

Enter the name of a state: Population of Maharashtra = 126000000
```

EXP 10

EXP10.cpp

ATM Machine Simulation & Exception Handling

Problem Statement: Develop a C++ program for simulation of a real-time application (ATM machine) with checking balance, withdrawing, and depositing money. Use try-catch blocks to handle potential exceptions such as insufficient funds and invalid inputs.
Core Concepts: Exception Hierarchy, Custom Exceptions (Arithmetic & IllegalArgument), Input Validation

SOURCE CODE:

```
// Experiment 10: ATM Machine Simulation & Exception Handling
#include <iostream>
#include <limits>
#include <stdexcept>

using namespace std;

// Custom exception for insufficient funds
class ArithmeticException : public runtime_error {
public:
  ArithmeticException(const string &message) : runtime_error(message) {}
};

// Custom exception for invalid input
class IllegalArgumentException : public invalid_argument {
public:
  IllegalArgumentException(const string &message) : invalid_argument(message) {}
};

class ATM {
private:
  double balance;

public:
  ATM(double initialBalance) { balance = initialBalance; }

  void checkBalance() {
    cout << "\nCurrent Account Balance: Rs. " << balance << endl;
  }

  void deposit(double amount) {
    if (amount ≤ 0) {
      throw IllegalArgumentException(
          "Deposit amount must be greater than zero.");
    }

    balance += amount;

    cout << "Rs. " << amount << " deposited successfully.\n";
    cout << "Updated Balance: Rs. " << balance << endl;
  }

  void withdraw(double amount) {
    if (amount ≤ 0) {
      throw IllegalArgumentException(
          "Withdrawal amount must be greater than zero.");
    }

    if (amount > balance) {
```

```
      throw ArithmeticException("Insufficient funds.");
    }

    balance -= amount;

    cout << "Rs. " << amount << " withdrawn successfully.\n";
    cout << "Remaining Balance: Rs. " << balance << endl;
  }
};

int main() {
  ATM atm(10000);
  int choice;
  double amount;

  cout << "=====================================\n";
  cout << "        ATM MACHINE SIMULATION      \n";
  cout << "=====================================\n";

  while (true) {
    cout << "\n------------ ATM MENU ------------\n";
    cout << "1. Check Account Balance\n";
    cout << "2. Deposit Money\n";
    cout << "3. Withdraw Money\n";
    cout << "4. Exit\n";
    cout << "-------------------------------------\n";
    cout << "Enter your choice: ";

    try {
      cin >> choice;

      if (cin.fail()) {
        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(), '\n');

        throw IllegalArgumentException(
            "Invalid input. Enter a number from 1 to 4.");
      }

      switch (choice) {
      case 1:
        atm.checkBalance();
        break;

      case 2:
        cout << "Enter amount to deposit: ";
        cin >> amount;

        if (cin.fail()) {
          cin.clear();
          cin.ignore(numeric_limits<streamsize>::max(), '\n');

          throw IllegalArgumentException("Invalid deposit amount.");
        }

        atm.deposit(amount);
        break;

      case 3:
        cout << "Enter amount to withdraw: ";
        cin >> amount;

        if (cin.fail()) {
          cin.clear();
          cin.ignore(numeric_limits<streamsize>::max(), '\n');

          throw IllegalArgumentException("Invalid withdrawal amount.");
        }
```

```
        atm.withdraw(amount);
        break;

      case 4:
        cout << "\nThank you for using the ATM.\n";
        return 0;

      default:
        throw IllegalArgumentException("Invalid choice. Please select 1 to 4.");
      }
    } catch (const ArithmeticException &e) {
      cout << "\nArithmetic Exception: " << e.what() << endl;
    } catch (const IllegalArgumentException &e) {
      cout << "\nIllegal Argument Exception: " << e.what() << endl;
    } catch (const exception &e) {
      cout << "\nException: " << e.what() << endl;
    }

    // Finally equivalent in C++
    cout << "\nTransaction completed.\n";
    cout << "ATM is ready for the next transaction.\n";
  }

  return 0;
}
```

TEST INPUT:

> 1 (Check Balance) -> 2 (Deposit 5000) -> 3 (Withdraw 3000) -> 4 (Exit)

EXECUTION OUTPUT:

Terminal — ./a.out

```
═══════════════════════════════════
        ATM MACHINE SIMULATION
═══════════════════════════════════

──────────── ATM MENU ────────────
1. Check Account Balance
2. Deposit Money
3. Withdraw Money
4. Exit
──────────────────────────────────
Enter your choice:
Current Account Balance: Rs. 10000

Transaction completed.
ATM is ready for the next transaction.

──────────── ATM MENU ────────────
1. Check Account Balance
2. Deposit Money
3. Withdraw Money
4. Exit
──────────────────────────────────
Enter your choice: Enter amount to deposit: Rs. 5000 deposited successfully.
Updated Balance: Rs. 15000

Transaction completed.
ATM is ready for the next transaction.
```

```
———————— ATM MENU ————————
1. Check Account Balance
2. Deposit Money
3. Withdraw Money
4. Exit
————————————————————————————
Enter your choice: Enter amount to withdraw: Rs. 3000 withdrawn successfully.
Remaining Balance: Rs. 12000

Transaction completed.
ATM is ready for the next transaction.

———————— ATM MENU ————————
1. Check Account Balance
2. Deposit Money
3. Withdraw Money
4. Exit
————————————————————————————
Enter your choice:
Thank you for using the ATM.
```